

Time Series Analysis with Cryptocurrency

Applying data analytics and machine learning to predict cryptocurrency price movements



Project Overview

Cryptocurrency markets are highly volatile and unpredictable, making informed decisions challenging for investors and traders. Traditional financial models often fail to capture the complex patterns in cryptocurrency price movements.

This project leverages historical data and machine learning models to identify patterns, trends, and potential future price movements in Bitcoin.



Project Objectives

01

Data Collection

Collect and analyze historical cryptocurrency data

02

Data Preprocessing

Perform cleaning and exploratory data analysis

03

Model Implementation

Implement time series forecasting models

04

Dashboard Development

Develop interactive dashboard using Streamlit

05

Market Insights

Provide insights into trends and volatility

Data Collection & Preprocessing



Dataset Source

Collected using yfinance API

- Bitcoin (BTC-USD)
- Historical data from 2010-2025
- Daily price data



Preprocessing Steps

Ensured clean, consistent data

- Removed missing values
- Converted date formatting
- Sorted chronologically
- Dropped duplicates



Features

Key price indicators

- Open, High, Low, Close
- Trading volume
- Date timestamps

Exploratory Data Analysis

Key Insights

- Bitcoin prices show high volatility
- Significant long-term growth observed
- Frequent fluctuations indicate risk and opportunity

Tools Used

Matplotlib for basic plots and **Plotly** for interactive visualization



Time Series Forecasting Models



ARIMA

AutoRegressive Integrated Moving Average

- Statistical model for time series
- Captures linear patterns
- Effective for short-term predictions



LSTM

Long Short-Term Memory

- Deep learning model based on RNN
- Captures long-term dependencies
- Handles non-linear patterns



Prophet

Facebook's forecasting tool

- Handles trend and seasonality
- Robust with real-world data
- Easy to use

Volatility & Sentiment Analysis

Volatility Analysis

Measured degree of price variation using:

- Rolling Mean (Moving Average)
- Rolling Standard Deviation

Helps identify market risk and understand price fluctuations

Sentiment Analysis

Basic sentiment module implemented with three categories:

- Positive 😊
- Negative 😞
- Neutral 😐

Can be extended using NLP for real-time news analysis

Interactive Dashboard



Streamlit Framework

Built using Streamlit for web interface and Plotly for interactive charts



Real-Time Visualization

Interactive line charts with KPI metrics showing current, highest, and lowest prices



Dataset Features

Dataset preview, statistical summary, volatility analysis, and market sentiment display



Results & Applications

Successfully Implemented

- Collected and processed cryptocurrency data
- Visualized trends effectively
- Multiple forecasting models
- Interactive dashboard

Practical Applications

- Assists traders in informed decisions
- Helps investors analyze trends
- Financial analytics and research
- Real-time trading systems

Future Scope & Conclusion

Future Enhancements

- Real-time sentiment analysis using NLP
- Support for multiple cryptocurrencies
- Cloud platform deployment
- Advanced deep learning models
- Integration with trading systems

Conclusion

This project demonstrates the application of data analytics and machine learning in cryptocurrency analysis. By combining data preprocessing, visualization, and forecasting models, it provides a comprehensive approach to understanding market behavior.

The interactive dashboard enhances usability and highlights the importance of data-driven decision-making in financial markets.